PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc (Food Processing and Preservation)

as per NEP 2020

w.e.f session 2024-25

Course Outline for B.Voc (Food Processing and Preservation) as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 101	Core Discipline	Fundamentals of Food Science	2	2	4	40	10	40	10	100
BFP 102	Core Discipline	Bakery and Confectionery Technology	2	2	4	40	10	40	10	100
BFP 103	Core Discipline	Quality Assurance	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BFP 201	Core Discipline	Food Preservation Technology	2	2	4	40	10	40	10	100
BFP 202	Core Discipline	Food Plant Layout	2	2	4	40	10	40	10	100
BFP 203	Core Discipline	Dairy Science and Technology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Bakery Technician

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Food Processing and Preservation = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (FOOD PROCESSING AND PRESERVATION) (SEMESTER I & II)

PROGRAMME OUTCOMES

PO1. Understanding of the concepts, principles of working of the Food Industry.

PO2. To understand the different preservation techniques of foods by heat processing, by low temperature, by removal of water and by using salt and sugar or by other relevant methods.

PO3. To identify the various microorganisms pathogenic or non-pathogenic and trace the cause of food contamination.

PO4. Describe the spoilage and deterioration mechanisms happening in foods and food products and to apply methods to control microbial activities prolonging shelf life, assuring it fit for human consumption.

PO5. To plan and design a food plant layout in accordance with the type of establishment (small, medium or large) in a specific area.

PO6. To understand Food laws and Food Safety regulations in India and worldwide, this will be accompanied by Safety and hygienic practices to be followed in a food plant layout.

PO7. To develop ability to apply principles of food engineering in industry.

PO8. Process and prepare food products which could be brought in the concurrent food market for sale.

PO9. To develop awareness of the marketing strategies, management hierarchy involved in a food processing plant.

Semester I

	COURSE TITLE	FUNDAMENTALS OF FOOD SCIENCE
1	**Course Code**	BFP 101
2	**Typology of Course**	Core Discipline
3	**Course Duration/LP (Lecture/Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week)(1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Bakery Technician
7	**Course Objectives**	To understand the importance of nutrients in our daily diet and to formulate nutritionally enriched food products.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To develop proficiency skill in producing different nutritious food products. CO2. Make different processed food products with quality assurance. CO3.Understand the structure, composition, and nutritional quality of foods.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction to Food Science: Definition & Scope. Basic concept of Food: Nutrient, Nutrition, Classification of Nutrients: Macronutrients, Micronutrients; dietary guidelines. Concept of food groups.

UNIT– II

Constituents of food - Definition, occurrence, properties, significance: Carbohydrates, Proteins, Fats and oils; Nutritive aspects. Protein quality. Bioavailability of nutrients. Vitamins. Minerals. Fiber. Water.

UNIT– III

Food and nutrition - Energy metabolism - BMR. Recommended dietary allowances, Balanced diet for different age groups (Infancy to old age). Stability of nutrients. Role of nutrients in health and disease prevention.

UNIT-IV

Food Processing: Unit operations in food industries: Materials handling, cleaning, separating, grinding, mixing, heat exchanging, forming, packaging.

Practical Work:

1. Demonstration of basic laboratory equipments used in food processing labs.
2. Demonstration of chemistry of candy making and the effect of temperature on the texture of candies.
3. Calculation of BMR
4. Enrichment and fortification of routine diet.
5. Demonstration of unit operation in food processing: mixing, separating, heat exchanging,packaging.

12 **Suggested Resources for Reading**

1. Food Science by B. Srilalshmi, New Age International Private Limited; Sixth Edition,2015.
2. Food: Facts and Principles by N. Shakuntala Manay, M. Shadaksharaswamy, New Age International Private Limited; Fourth Edition, 2020.
3. Food Science by Joseph H. Hotchkiss and Norman N Potter, Aspen Publishers Inc.,U.S,Fifth edition,2007
4. Essentials of Food Science by Vickie A. Vaclavik, Elizabeth W. Christian, Springer Publications, Fourth Edition,2013
5. Food Science by Sunetra Roday, Oxford Publication, Third Edition, 2018

Note: Latest edition of text books may be used.

	COURSE TITLE	BAKERY AND CONFECTIONARY TECHNOLOGY
1	**Course Code**	BFP 102
2	**Typology of Course**	Core Discipline
3	**Course Duration/LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week)(1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Bakery technician.
7	**Course Objectives**	To impart basic knowledge of scope of bakery and confectionery and learn about bakingprocess.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarizing the students with different equipment and ingredients used in bakeryunit and their functions. CO2. The students will be able to identify various bakery and confectionery productsCO3.The learners will develop fundamental baking techniques. CO4. Work effectively as a member of a team and learn entrepreneurial skills for self-reliance. CO5. The students will learn the basics of setting up a bakery unit.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions, Case Studies etc.
10	**Instructions for paper setters andcandidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions fromeach unit, including Question No. 1 (compulsory) of short answer type coveringthe whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction- Structure of wheat grain, composition and milling process, Ingredients used in bakery process and their role, types of flour, leavening agents and its types.

UNIT– II

Baking Methods: Various Methods of Bread Making: (Straight dough method, Sponge dough method.) Various Cake making, methods (Sugar batter, Flour batter method), Pastry Making: Principle and types

UNIT– III

Bakery Technology- Sensory attributes of bakery products and factors affecting it, Bread faults and remedies, Various equipments used in bakery industry- Small and large equipments and their role

UNIT-IV

Confectionery- Types of Confectionery, raw materials and processing of toffee, Fondant, hard boiled sweets. Sugar Panned Confections and their types.

Practical Work:

1. Preparation of White Bread.
2. Preparation of Fruit cake.
3. Studying the effect of temperature on the process of biscuit making.
4. Preparation of garlic bread and sensory analysis.
5. Preparation of icing of cakes.
6. To estimate WAP of the flour sample
7. To determine gluten content in the dough

12 **Suggested Resources for Reading**

1. Textbook of bakery and confectionery, by Yogambal Ashok Kumar
2. Bakery products science and Technology by Weibiao Zhou, Wiley Blackwell, second edition
3. A Professional Text to Bakery and Confectionary by John Kingslee
4. Bakery-I: Student Handbook and Practical Manual- CBSE.
5. Bread: A Baker's Book of Techniques and Recipes by Jeffrey Hamelman-Wiley, 2012.
6. Confectionery Science and Technology by Richard W. Hartel, Joachim H. von Elbe Randy Hofberger,Springer, 2017

Note: Latest edition of text books may be used.

	COURSE TITLE	QUALITY ASSURANCE
1	**Course Code**	BFP 103
2	**Typology ofCourse**	Core Discipline
3	**Course Duration/LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week)(1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marksand Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of thedevelopments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Bakery Technician
7	**Course Objectives**	To develop a broad understanding of the principles of modern quality management systems; to understand objective and subjective factors that contribute to the quality of foods or food-related services within the food value chain.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Illustrate the importance of food quality standards and of Quality Assurance. CO2.Understand the principles of Total Quality Management to evaluate and select between alternative strategies in designing a quality management system. CO3.Explain the application of food quality and food safety system. CO4.Understand the role of food quality assurance system standards in India.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for Paper Setters and Candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Objectives, importance and functions of quality control. Principles of food quality control and quality assurance, quality control and assessment in food materials used in the industry.

UNIT– II

Total quality management; Good Manufacturing Practices, Good Hygienic Practices, Good Laboratory Practices, Quality Circles, general awareness and role of management practices in quality control.

UNIT– III

Food laws and standards: Food regulations and standards in Indian scenario, concepts of Codex Alimentarius, HACCP, USFDA, ISO 9000 series etc. Food standards and safety Act: salient provisions and prospects.

UNIT-IV

Sensory quality evaluation: Objectives, type of food panels, characteristics of panel member, layout of sensory evaluation laboratory, Types of Sensory Testing: difference tests, Rating tests, Sensitivity test, Descriptive test. Chemical dimension of basic tastes.

Practical work

1. Techniques of quality assessment of different natural and processed foods.
2. Identification and ranking of food product attributes
3. Sensory methods for measuring food quality assessment of raw materials used and processed bakery products
4. Study of cleaning and sanitizers used in pre- and post-operative processes in bakery industry
5. Documentation of details of baking ingredients, process and finished products used in baking industry.

12 **Suggested Resources for Reading**

1. Food Quality Assurance Principles and Practices by Inteaz Alli, CRC Press, 2004.
2. Quality Assurance for the Food Industry. A Practical Approach by J Andres Vasconcellos, CRC Press, 2003
3. Safety And Quality Assurance in Food Supply Chain: Emerging Technologies and challenges by K.P. Sudheer, Bindu Lakshmanan New India Publishing Agency (2021)
4. Food Science by Sunetra Roday, Oxford publication, Third edition, 2018
5. Sensory Evaluation Techniques by Morten C. Meilgaard, B. Thomas Carr, GailVance Civille, CRC Press LLC, Fourth edition, 2006.
6. Principles of Sensory Evaluation of food by Amerine, Pangborn & Roessler, Academic Press, London. 2013

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	FOOD PRESERVATION TECHNOLOGY
1	**Course Code**	BFP 201
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in thesubject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6	**Job Role**	Bakery Technician
7	**Course Objectives**	The goal of this course is to provide students with a fundamental understanding of food preservation
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Will be able to comprehend the reason for food spoilage. CO2. Will be able to comprehend the principles of preventing contamination and theremoval of microorganisms methods used for food safety. CO3. Students will be able to understand various heat preservation techniques, low temperature techniques.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type coveringthe whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction: scope and benefit of industrial food preservation, perishable, non-perishable food, causes of food spoilage. Preservation by salt and sugar – Principle, method, equipment and effect on food quality

UNIT– II

Thermal processing methods of preservation – Canning, blanching, pasteurization, sterilization, evaporation, Preservation by fermentation- Definition, advantages, disadvantages, types, equipments

UNIT– III

Food preservation by use of low temperature – Principle, processes and effect on quality. Preservation by drying, dehydration and concentration – Principle, Methods, equipment and effect on quality of food.

UNIT-IV

Recent methods in preservation: Pulsed electric field processing, High pressure processing, Processing using ultrasound, Microwave heating, Ohmic and infrared heating

Practical work:

1. Demonstration of various machineries used in food processing.
2. Demonstration on the effect of blanching on the quality of foods.
3. Preservation of food by high concentration of sugar i.e. preparation of jam
4. Preservation of food by using salt
5. Preservation of milk by condensation/concentration
6. To Demonstrate the Preservation of Fruits by drying\

12 **Suggested Resources for Reading**

1. Food Science- Potter, CBS publishers.
2. Food Processing and Preservation- Subbulaksmi G., and Udipi S
3. Principles of Food Science, Vol. II- G. Borgstron, Mc. Millan Co. Ltd. London
4. _Principles of food preservation Part I& II- Owen R. Fenemma.
5. Technology of Food Preservation - N.W. Desroiser and N.W. Desrosier

Note: Latest edition of text books may be used.

	COURSE TITLE	FOOD PLANT LAYOUT
1	**Course Code**	BFP 202
2	**Typology ofCourse**	Core Discipline
3	**Course Duration/LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks andTime**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in thesubject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6	**Job Role**	Bakery Technician
7	**Course Objectives**	i) Exposure of the students to the basic setup of a food industry. ii) To make them conversant with the machinery and equipment in a food industry.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. understand concepts of plant layout. CO2. have knowledge on building, utilities in the plant. CO3. know the importance of proper food plant design and safety. CO4. evaluate and develop problem solving skills regarding food processing operationsthat can affect the quality of foods. CO5. know and implement good manufacturing practices.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions, Case Studies etc.
10	**Instructions for paper setters andcandidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type coveringthe whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Industrial plant layout and design: Basic concepts of Plant Layout with reference to food industries. Hygiene requirements in food plant layout and design.

UNIT– II

Plant Design: Design consideration for location of food plants. Basic understanding of equipment layout and ventilation in food processing plants. Preparation of flow sheets formaterial movement and utility consumption in food plants.

UNIT– III

Plant Layout: Preparation of a Plant Layout, Plant Layout problem, importance, objectives, classical types of layouts. Evaluation of layout. Advantages of good layout

UNIT-IV

Plant Layout Requirements: Common materials of construction of Food plant, building. Maintenance of Food Plant Building, Illumination and ventilation, Cleaning and sanitization, painting and color coding, Fly and insect control.

Practical Work:

Industrial visit and report making.

12 **Suggested Resources for Reading**

1. Hygienic Design of Food Factories, by John Holah, H. L. M. Lelieveld, Elsevier Publication, 2011
2. Practical Design, Construction and Operation of Food Facilities by J. Peter Clark, Academic Press Publishers.2008
3. Food Plant Economics by Zacharias B. Maroulis, George D. Saravacos, CRC Press Publishers. 2007
4. Food Plant Design by Antonio Lopez-Gomez, Gustavo V. Barbosa-Canovas, CRC Press Publishers. First edition, 2005

Note: Latest edition of text books may be used.

	COURSE TITLE	DAIRY SCIENCE AND TECHNOLOGY
1	**Course Code**	BFP 203
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week)(1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in thesubject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6	**Job Role**	Bakery Technician
7	**Course Objectives**	Students will learn about Principles of Dairy, processing of technologies tohandle physical aspects of fluid-based milk and its various products.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. In-depth knowledge of dairy production and processing techniques. CO2. Comprehensive understanding of dairy production, processing, and preservation techniques. CO3. Better understanding of the concepts underlying Dairy processing as wellas the reliability and caliber of Dairy products. CO4. Students will learn about technologies to handle physical aspects of fluid-based process CO5. Students will learn about various equipments present in advanced dairy
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type coveringthe whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction- Milk definition, composition of milk, nutritive value of milk, physico-chemical propertiesof milk constituents and physico- chemical properties of milk, factors affecting quality of milk.

UNIT– II

Market Milk- Manufacture, packaging and storage of standardized pasteurized milk, Various methods of pasteurization, Judging and grading of milk, Flavour defects in milk, their causes and Prevention,

UNIT– III

Special milks- Sterilized milk, Homogenized milk, Flavoured milks, Reconstituted/Rehydrated milk, Recombined milk, Toned milk, Double Toned milk
Fermented milk- Acidophilus milk, Bulgarian buttermilk, Kumiss and kefir.

UNIT-IV

Dairy Products - Composition and manufacturing of butter, butteroil, Ice cream and cheese

Practical Work:

1. To perform platform tests for milk analysis- analysis of milk for its organoleptic properties
2. To determine stability of milk using COB Test
3. To test milk quality by performing MBRT Test
4. To perform Alcohol test to assess raw milk quality
5. Sediment test to check for adulteration in milk sample
6. Determination of titratable acidity of milk
7. To determine Fat content by Gerber method

12 **Suggested Resources for Reading**

1. Outlines of Dairy Technology, by Sukumar De, Oxford University Press.
2. Milk Hygiene in Milk Production Processing and Distribution, F.A.O Publication.

Note: Latest edition of text books may be used.